

ENTERPRISE TECHNICAL ARCHITECTURE SPECIFICATION

ERP SOLUTION ARCHITECTURE & CYBERSECURITY DESIGN

High-Availability Multi-Tier Enterprise Resource Planning (ERP) System featuring React.js Custom Controls, Clean PHP MVC Backend, RabbitMQ Request Queueing, Offline SQLite/IndexedDB Storage Engine, Git Version Governance, and JetBackup 5 Incremental Disaster Recovery.

DOCUMENT REFERENCE
SL-POP-ERP-ARCH-001

SYSTEM RELEASE
Version 1.0 (Enterprise)

TARGET INFRASTRUCTURE
3-Tier Virtualized Server

SECURITY & BACKUP TIER
Zero-Trust / JetBackup 5
Incremental

1. Executive Summary & Core Architectural Principles

The Popular Auto Spare ERP Solution is engineered as an ultra-reliable, high-throughput, mission-critical enterprise system. It unifies multi-branch automotive spare parts cataloging, counter POS billing, warehouse inventory ledgering, and financial accounting across the Kingdom of Bahrain. The architectural philosophy prioritizes:

⚡ Sub-Second Latency

- Optimistic React.js UI state
- Redis distributed query caching
- Bespoke lightweight UI controls

📶 Offline Resilience

- IndexedDB local transaction log
- Native POS terminal SQLite DB
- Bi-directional delta auto-sync

🔄 Async Queueing

- RabbitMQ message broker
- Background ledger reconciliation
- Zero thread blocking on POS writes

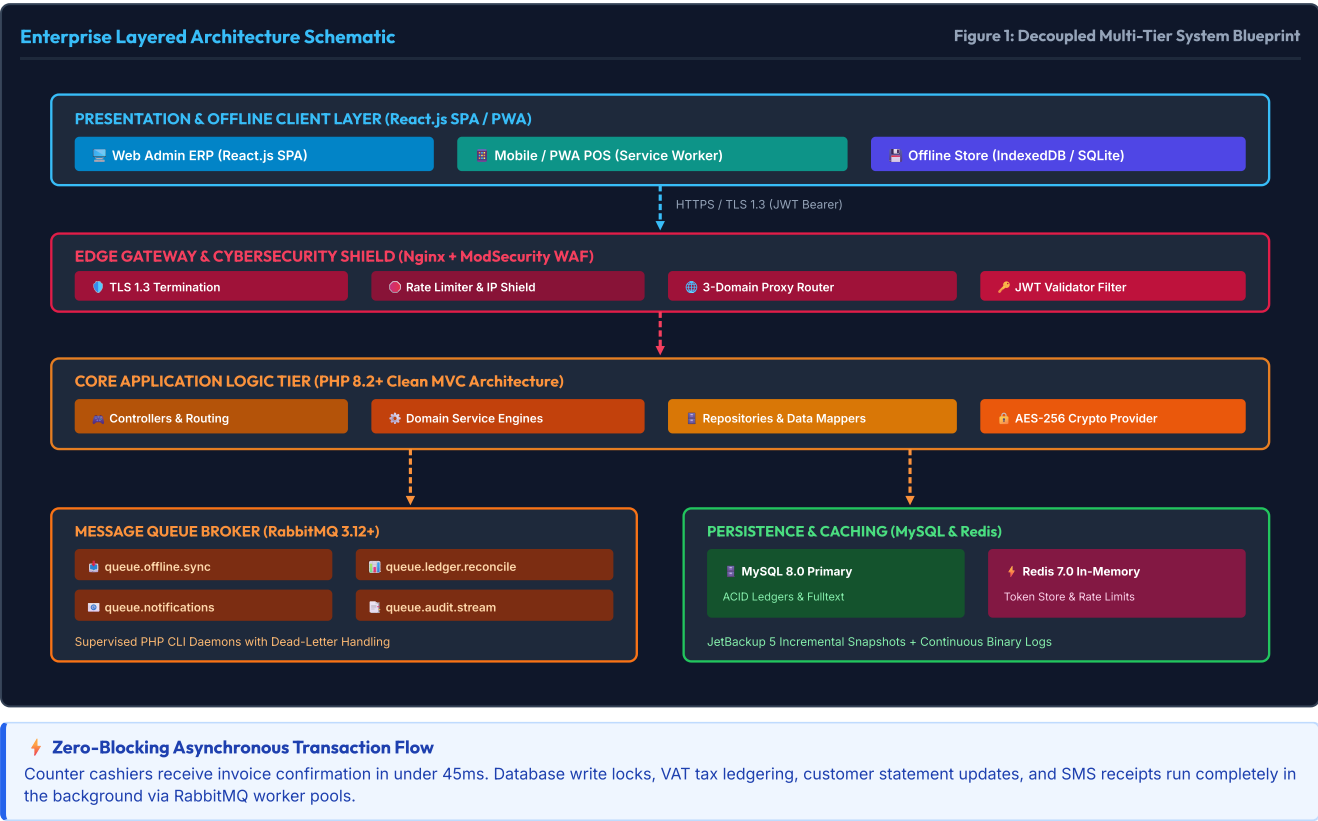
🔒 Git & JetBackup DR

- 3-Tier Git branching governance
- JetBackup 5 incremental snapshots
- Point-in-Time Recovery (PITR)

LAYER / TIER	CORE TECHNOLOGY	KEY LIBRARIES & DRIVERS	ARCHITECTURAL PURPOSE
Client Presentation	React.js 18.2+ (SPA & PWA)	Custom Virtual DOM Controls, Lucide Icons, Canvas API	Reactive, touch-optimized POS & Admin portal.
Local Offline Store	IndexedDB (PWA) / SQLite (Native)	Dexie.js 3.2+ Wrapper, SQLite3 Embedded C-Engine	Zero-latency offline billing & mutation outbox.
Edge & Gateway	Nginx 1.24+ Reverse Proxy	ModSecurity WAF, Let's Encrypt TLS 1.3, Brotli	SSL termination, DDoS mitigation, virtual host routing.
Application Backend	PHP 8.2+ Clean MVC Core	FastRoute, PHP-DI, Respect/Validation, PDO MySQL	Stateless REST APIs, domain business logic rules.
Message Queue Broker	RabbitMQ 3.12+ (Erlang OTP 26)	AMQP 0-9-1 Protocol, RabbitMQ Management Plugin	Async batch ingestion, sync payload processing.
Persistence & Cache	MySQL 8.0+ & Redis 7.0+	InnoDB ACID Engine, Redis In-Memory KV Cache	Master financial ledgers, token store & rate limits.
Backup & Recovery	JetBackup 5 + MySQL PITR	Dedicated NVMe Backup Volume, AES-256 S3 Sync	Incremental snapshots, RPO < 1h, RTO < 30m.

2. End-to-End System Architecture Schematic

The system is organized into a clean decoupled 5-tier architecture. High-frequency write requests from branch POS terminals and administrative web consoles pass through the Nginx Edge Gateway directly to the PHP MVC micro-kernel. Heavy reconciliation tasks and offline delta synchronizations are queued into RabbitMQ to guarantee maximum responsiveness.



⚡ Zero-Blocking Asynchronous Transaction Flow

Counter cashiers receive invoice confirmation in under 45ms. Database write locks, VAT tax ledgering, customer statement updates, and SMS receipts run completely in the background via RabbitMQ worker pools.

3. Frontend Architecture (React.js Bespoke Custom Controls)

The user interface is constructed using **custom-engineered high-performance React.js components** without bloated dependencies, ensuring high frame rates during counter checkout and instantaneous search over 100,000+ spare parts cross-references.

**Virtual DOM High-Speed DataGrid**

Renders 50,000+ automotive spare part SKUs using window virtualization (only rendering visible rows). Sub-16ms frame times, multi-column search by OEM Part #, brand, and vehicle chassis compatibility.

**POS Fast-Checkout Engine & Scanner**

Keyboard-first counter billing with **Automatic 1D/2D Barcode & QR Code Listener**. Intercepts hardware HID scanners globally with focus-bypass, decodes live camera streams via WebRTC frame analyzers, handles split payments, and outputs direct ESC/POS thermal receipts.

**Vector Digital Signature Canvas**

Bezier curve smoothing signature pad with embedded SHA-256 integrity watermark, capturing pressure and hardware timestamps for delivery notes, credit approvals, and milestone sign-offs.

**Offline Service Worker Cache**

Progressive Web App (PWA) cache engine storing application bundles and catalog master data. Emits network availability heartbeats and initiates transactional background synchronization queues.

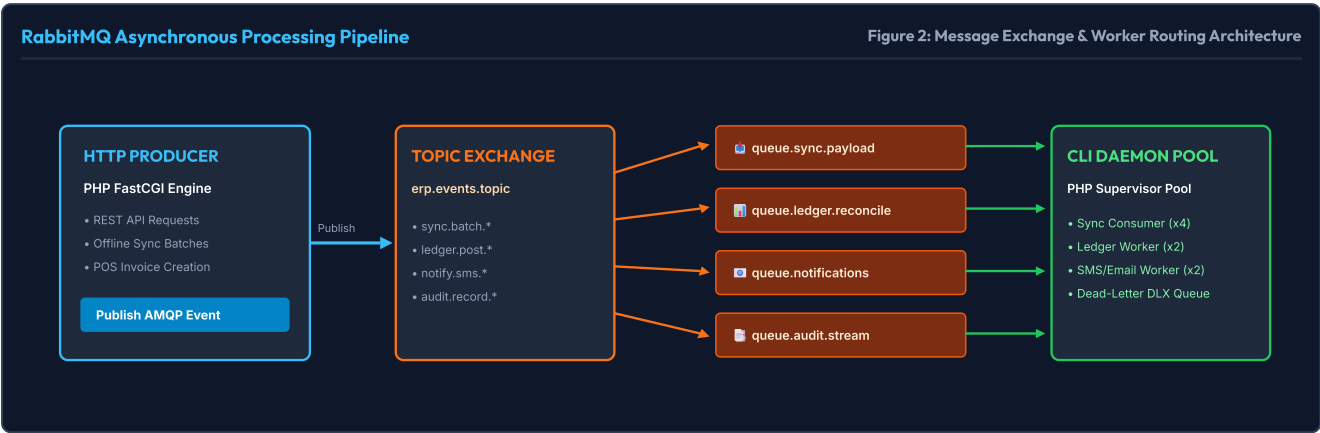
4. Backend Architecture (PHP 8.2+ Clean MVC Core)

The backend enforces strict separation between HTTP transport, domain business logic, and database access. Controllers remain strictly thin, delegating all processing to transactional Domain Services.

LAYER	COMPONENT NAME	CORE RESPONSIBILITIES	ENFORCEMENT
1. HTTP Gateway	Nginx & Global Middleware	Request parsing, IP filtering, rate-limiting, CORS, Brotli compression.	WAF & TLS 1.3 Termination
2. Middleware	Auth & Schema Handlers	JWT token verification, permission bitmask checks, input payload sanitization.	Replay token rejection
3. Controllers	Thin MVC Controllers	Input mapping, Service delegation, JSON:API structured output formatting.	Zero raw SQL allowed
4. Domain Services	Pure Business Engines	VAT calculation, double-entry inventory ledger updates, discount validation.	Atomic DB transactions
5. Repositories	Data Access Layer	Abstracting SQL data access via strict PDO Prepared Statements, Redis cache.	100% SQLi immunity
6. Message Queue	AMQP RabbitMQ Client	Publishing async events (Sync, Ledger Rollup, SMS, Audit Stream) to broker.	Persistent delivery mode

5. Asynchronous Request Queueing & RabbitMQ Integration

To prevent server lock contention and eliminate checkout lag during peak store hours, the ERP implements **RabbitMQ (AMQP 0-9-1)** as a high-throughput message broker. Time-consuming operations are offloaded from HTTP request cycles into dedicated background worker queues.



- 1 Zero-Blocking Ingestion

POS checkout responses return in < 45ms to the cashier. Complex multi-table inventory ledger updates and cost calculations are published to 'queue.ledger.reconcile' and executed asynchronously.
- 2 Backpressure & Throttling

If multiple branches upload large offline transaction batches simultaneously upon internet reconnection, RabbitMQ buffers messages smoothly, preventing database lock exhaustion or CPU spikes.
- 3 Dead Letter Exchange (DLX)

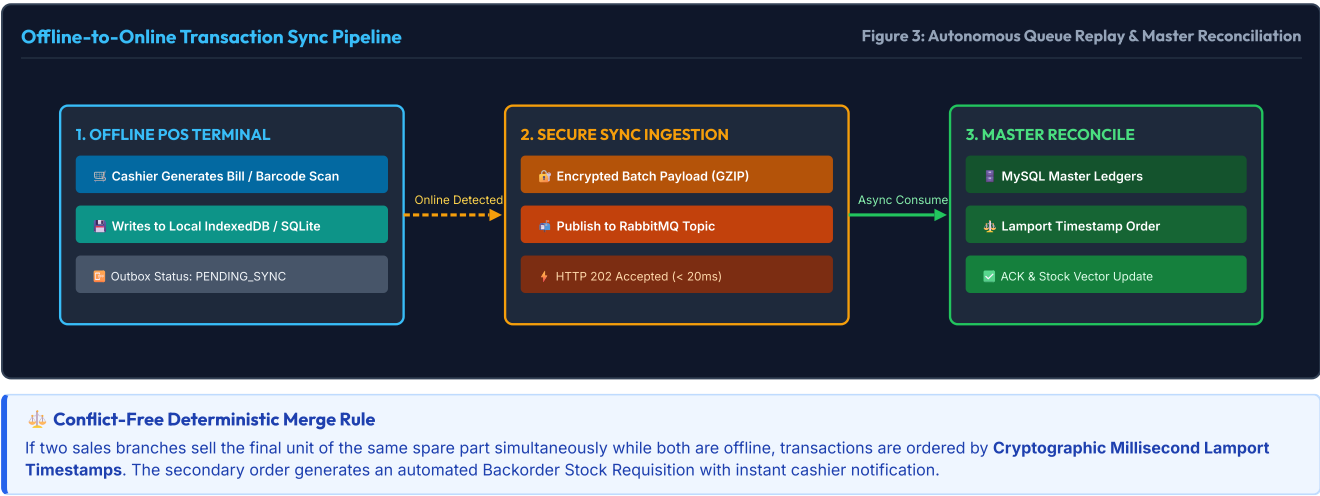
Failed tasks undergo exponential retry (3 attempts). Persistent anomalies are safely routed to a Dead Letter Queue ('dlq.failed.events') with real-time alerts to the Super Admin.

6. Offline-First Architecture & Bi-Directional Auto-Sync Engine

Automotive spare parts sales counters must operate continuously without interruption, even during telecom outages, fiber cuts, or local Wi-Fi disruptions. The ERP implements an **Autonomous Offline Storage & Transaction Queue** that seamlessly transitions between offline and online states.

Client Storage Engine
IndexedDB (Dexie.js): Stores complete branch stock catalogs, customer accounts, and pending invoice outbox locally in browser memory.
SQLite Engine (Native Electron): Standalone POS counters maintain a local relational SQLite database with full ACID transactions.

Auto-Sync Coordinator
Heartbeat Monitor: Pings `/api/health` every 5 seconds.
Delta Replay Engine: Packages pending mutations into encrypted JSON batches and flushes sequentially to RabbitMQ upon reconnection with deterministic conflict resolution.



7. Cybersecurity Architecture, Zero-Trust Model & Cryptography

The Popular Auto Spare ERP follows a strict **Zero-Trust Security Architecture**. Every single API request is authenticated, cryptographically signed, rate-limited, and audited against replay attacks.

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Rotating JWT Authentication

- Access Token (15m):** Short-lived RS256 signed.
- Refresh Token (30d):** Stored in 'HttpOnly' Secure cookie.
- Sliding JTI Whitelist:** Redis revokes stolen tokens immediately.

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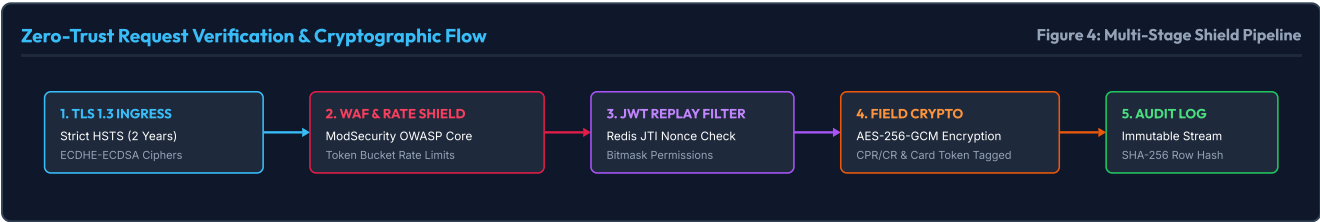
Data-at-Rest Field Crypto

- AES-256-GCM:** Encrypts CPR, commercial records, BenefitPay tokens.
- Argon2id Hashing:** 64MB memory cost for employee credentials.
- Per-Tenant Salts:** Isolated encryption vectors.

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Tamper-Proof Audit Trail

- SHA-256 Block Chaining:** Financial ledger lines form a hash chain.
- IP & Geo-Fencing:** Captures device fingerprints and login IPs.
- Immutable Logs:** Write-once audit tables preventing edits.



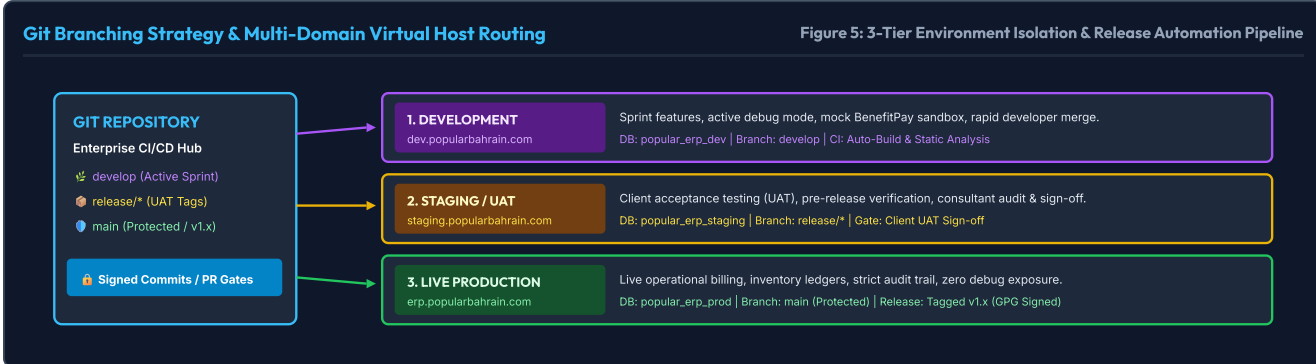
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Comprehensive OWASP Top 10 Mitigation Matrix

- Injection Attacks (SQLi/Command):** 100% strictly parameterized PDO prepared statements; zero string concatenation in queries.
- Cross-Site Scripting (XSS):** Automatic HTML entity encoding via Twig/React JSX, strict 'Content-Security-Policy' headers, and input DOMPurify sanitization.
- Cross-Site Request Forgery (CSRF):** Anti-CSRF cryptographic token validation on state-changing non-GET endpoints combined with 'SameSite=Strict' cookies.
- Cryptographic Failures:** Passwords hashed using Argon2id / Bcrypt (cost factor 12); TLS 1.3 encryption for all data in transit with forward secrecy.
- Brute Force Protection:** Redis-backed sliding-window token bucket algorithm enforcing maximum 5 failed attempts per 15 minutes per IP/User.

8. 3-Domain Multi-Environment Topology & Git Version Governance

To ensure zero downtime, complete isolation of experimental engineering, and rigorous client User Acceptance Testing (UAT), the ERP infrastructure implements **strict Git branching governance** mapped directly to **three isolated domain environments**:



Branch Protection Controls

'main' and 'staging' branches are **strictly protected**. Direct commits and force pushes ('git push --force') are completely prohibited. All code merges require mandatory Pull Request (PR) peer review and approval.



Automated CI Quality Gates

Every pull request triggers automated CI pipelines: PHPStan Level 8 static analysis, ESLint syntax validations, security dependency audit (composer/npm audit), and zero-breaking-change database migration dry-runs.



Instant Rollback Tags

Every production deployment creates an immutable Git semantic release tag ('v1.0.1', 'v1.0.2'). In the event of an unforeseen regression, the release pipeline performs an automated rollback in **< 60 seconds**.

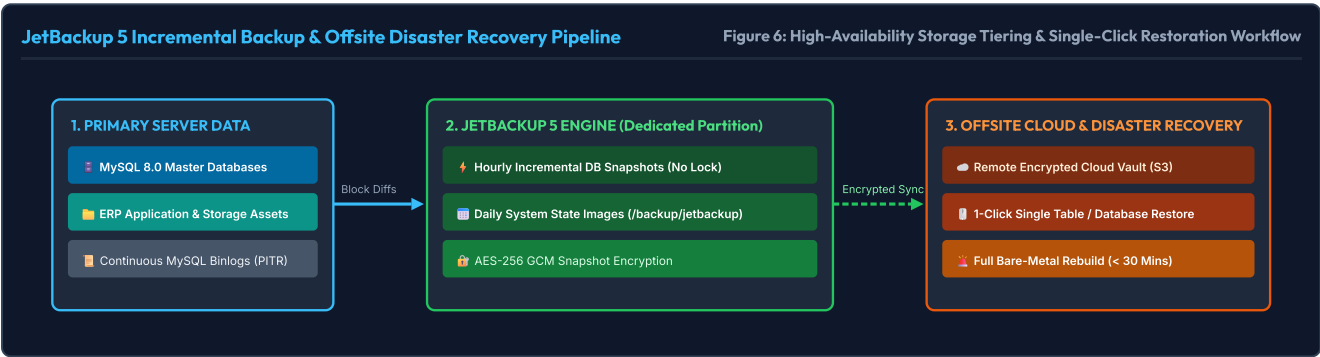
9. Enterprise Data Backup Architecture & JetBackup 5 Incremental Engine

To ensure enterprise business continuity, complete protection against data corruption, hardware failure, or ransomware, the server infrastructure is provisioned with **dedicated backup storage partitions** and the industry-standard **JetBackup 5 Incremental Backup Utility**.

Dedicated Disk Partition
Quota-Isolated Storage: A high-performance secondary NVMe mount (`/backup/jetbackup`) is isolated from the root OS and database storage. This guarantees that backup processing never starves ERP application disk space.

JetBackup Incremental Process
Block-Level Change Tracking: Only modified disk blocks and incremental database logs are captured after initial seeding. This reduces CPU/IO load by **>85%** and completely eliminates table locking during business hours.

MySQL Point-in-Time Recovery
Continuous Binary Logs (PITR): MySQL binary logging (`'binlog'`) runs continuously, enabling database restoration to the **exact second** prior to any accidental deletion or corruption event.



METRIC / OBJECTIVE	ARCHITECTURAL TARGET	IMPLEMENTATION MECHANISM	VERIFICATION SCHEDULE
RPO (Recovery Point)	< 1 Hour (Incremental) < 1 Min (with PITR)	JetBackup hourly delta snapshots + continuous MySQL binary log streaming.	Continuous automated check
RTO (Recovery Time)	< 30 Minutes (Full Server) < 5 Minutes (Database)	Direct JetBackup 1-click snapshot restore without manual file extraction.	Quarterly simulated drill
Retention Policy	24h Hourly • 30d Daily • 12m Monthly	Automated rolling grandfather-father-son backup rotation with auto-pruning.	Daily automated audit
Disaster Offsite Vault	100% Offsite Mirroring	Automated secondary sync to encrypted cloud object storage (AWS S3 / Wasabi).	Nightly offsite checksum check

Step-by-Step Disaster Recovery Runbook
In the event of total server hardware failure or data corruption: **(1)** Provision replacement VM instance → **(2)** Mount JetBackup storage partition → **(3)** Trigger 1-Click JetBackup snapshot restoration → **(4)** Replay continuous MySQL binary logs to exact point of failure → **(5)** Run automated ledger SHA-256 integrity hash verification → **(6)** Switch DNS / Nginx traffic back online.

This Technical Architecture Specification serves as the foundational engineering baseline for the Popular Auto Spare ERP Solution. It formalizes all performance benchmarks, queueing pipelines, offline auto-sync engines, cybersecurity layers, Git version controls, and JetBackup disaster recovery procedures across the Kingdom of Bahrain.

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